

Linux File System & Directory Structure

Simple Technical Notes for Students — Concepts, Commands & Key Definitions

1. What is a File System?

A **file system** is the method used by an operating system to store, organize, and manage data on a disk or server — including files, directories, and their related metadata — in a structured way.

Simple Analogy: Imagine 10,000 books lying on the floor of a room. Finding one specific book would be very difficult. But if the same books are organized properly on labeled shelves in a library, you can find any book easily. A file system does the same job for data on a computer — it organizes data so it can be accessed quickly and reliably.

Key Point: File system = Structured storage + Easy management of data.

2. The Root Directory (/)

In Linux, everything starts from the **root directory**, represented by a forward slash (/). It has no name of its own but is called the **parent directory** of the entire file system hierarchy. All other files and directories exist under it.

Command to view root directory contents:

```
cd /  
ls
```

The overall structure is similar across most Linux distributions (CentOS, Ubuntu, etc.), though minor differences may exist since each distribution can customize it slightly.

3. File Extensions

Just like Windows uses extensions such as **.docx** or **.pdf** to indicate a file's format, Linux files may also carry extensions, though Linux does not strictly require them to determine file type (Linux mainly relies on file content/metadata, not the extension).

4. Types of File Systems

Different file systems manage data differently. The most common types used in Linux are:

File System	Full Form / Meaning	Commonly Used In
ext3 / ext4	Extended File System (3rd / 4th Gen)	Ubuntu and Debian-based distros
XFS	Extents File System	CentOS / RHEL-based distros
Btrfs	B-Tree File System	openSUSE, Fedora (advanced use)
FAT	File Allocation Table	USB drives, older/Windows systems

Commands to Check Which File System a Server Is Using:

```
df -Th | lsblk -f | mount | grep <partition>
```

Running **df -Th** shows all partitions, their available space, and the type of file system used by each. For example, temporary and virtual partitions may show **tmpfs** or **vfat**, while the main partition (e.g. CentOS) shows **xfs**, and on Ubuntu it typically shows **ext4**.

XFS vs EXT4 — Key Differences

Feature	XFS	EXT4 (Extended 4)
Best suited for	Large files and large volumes	Wide range of file sizes
Performance	Superior performance and scalability for big data	Performs well but less efficient for extremely large files
Ideal use case	High throughput, parallel processing workloads	General-purpose systems

Conclusion: For high throughput and extensive parallel processing, XFS is typically more effective than EXT4. This is why CentOS/RHEL (Red Hat-based systems), popular in enterprise environments, commonly use XFS by default.

5. Concept of Inode

Basic commands to view file details:

```
ls -l    → shows size, creation/modification date, permissions
ls -li   → additionally shows the Inode Number
```

An **Inode Number** is a unique number associated with every file or directory when it is created. It represents the location of the memory block (real disk space) where the file's actual data is stored.

Example: If you create a file with content "a" inside it, that data ("a") must be physically stored somewhere on the disk. The Inode keeps track of exactly where that data block is located.

Key Point: The file system uses the Inode Number, which contains a pointer to the data block where the actual file data is stored, to track and retrieve file data efficiently.

6. File System Hierarchy — Important Directories

The root directory (/) is the starting point of the entire file system hierarchy. Below is an explanation of the most important default directories and their purpose.

/bin

Contains essential binary/executable command files used by all users (e.g. cd, ls, mkdir). Every basic command has its own executable file here.

/etc

Stores configuration files for the entire system — networking (DNS, hosts), user management, package/service configuration. Example: user accounts info is in /etc/passwd; DNS server configuration is in /etc/resolv.conf.

/home

Contains personal directories for all regular (non-root) users. Each user gets a dedicated folder here to store their own files and scripts. Users normally cannot access another user's home folder due to permission restrictions.

/root

The special home directory reserved only for the root (super) user — separate from the normal /home directory since root is a privileged, special user.

/var

Important from a troubleshooting point of view. Contains variable data such as cache, databases, and — most importantly — system logs, stored under /var/log.

/var/log

Contains different types of log files, for example: 'boot' logs (system boot process), 'secure' logs (login/access attempts), 'maillog' (mail server activity), service-specific folders (e.g. httpd for web servers), and 'messages' — the most general log file containing overall system activity.

/boot

Contains the necessary files required to boot (start up) the system or machine.

/dev

Contains files representing hardware devices connected to or used by the system (disks, partitions, ports, etc.). For example, lsblk shows hardware/partition details linked to files in this directory.

/lib and /lib64

Contain shared libraries used by different processes and operations running on the system.

/media

Used for automatically mounting removable/external devices (USB drives, CDs, etc.) when they are attached to the system.

/mnt

Also called the 'mount' directory. Similar to /media but generally used for manually and temporarily mounting devices — for example, attaching an external hard disk just to take a backup, then unmounting it once the task is complete.

/opt

Used for installing third-party or optional application software that is not part of the default distribution package.

/proc

A virtual and dynamic directory that provides information about running processes and hardware. Example: 'cat /proc/cpuinfo' shows CPU details; 'cat /proc/meminfo' shows total, free, and available memory.

/run

A temporary file system that stores transient state files, such as process IDs (PIDs) and lock files, related to currently running processes and applications.

/sbin

Similar to /bin but contains executable commands mainly needed by system administrators (e.g. fdisk) rather than regular users.

/srv

Also called the 'serve' directory. Used to store data that is being served through different services — for example, files shared via FTP or web pages served over HTTP.

/sys

Similar to /proc — another virtual file system that provides an interface to the kernel. Contains information and settings about the system's devices, drivers, and kernel features (e.g. 'cat /sys/class/block/sda/size' shows a partition's size).

/usr

Stands for 'Unix System Resources' (not 'user'). Used for user-level program functionality — contains its own bin, sbin, lib, include (header files), and local sub-directories. Keeping these resources at this level means everyday user-level operations do not directly impact core system files.

7. Quick Revision Table

Directory	One-Line Meaning
/bin	Contains essential binary/executable command files used by all users (e.
/etc	Stores configuration files for the entire system — networking (DNS, hosts), user management, package/service
/home	Contains personal directories for all regular (non-root) users.
/root	The special home directory reserved only for the root (super) user — separate from the normal /home directory
/var	Important from a troubleshooting point of view.
/var/log	Contains different types of log files, for example: 'boot' logs (system boot process), 'secure' logs (login/access
/boot	Contains the necessary files required to boot (start up) the system or machine.
/dev	Contains files representing hardware devices connected to or used by the system (disks, partitions, ports, etc
/lib and /lib64	Contain shared libraries used by different processes and operations running on the system.
/media	Used for automatically mounting removable/external devices (USB drives, CDs, etc.
/mnt	Also called the 'mount' directory.
/opt	Used for installing third-party or optional application software that is not part of the default distribution packag
/proc	A virtual and dynamic directory that provides information about running processes and hardware.
/run	A temporary file system that stores transient state files, such as process IDs (PIDs) and lock files, related to c
/sbin	Similar to /bin but contains executable commands mainly needed by system administrators (e.
/srv	Also called the 'serve' directory.
/sys	Similar to /proc — another virtual file system that provides an interface to the kernel.
/usr	Stands for 'Unix System Resources' (not 'user').

8. Key Takeaways for Students

- A file system organizes and manages all data on a server/machine in a structured way.
- Common Linux file systems: ext3, ext4 (Ubuntu/Debian), XFS (CentOS/RHEL), Btrfs, FAT.
- Check file system type using: `df -Th`, `lsblk -f`, or `mount`.
- XFS is better for very large files and high-throughput/parallel workloads; EXT4 is a solid general-purpose choice.
- Every file/directory has a unique Inode Number pointing to where its data is stored on disk.
- The entire hierarchy starts at root (/); everything else branches from it.
- /etc = configuration files, /var/log = system logs, /home = user files, /root = root user's home, /bin & /sbin = commands, /proc & /sys = live system/kernel info, /mnt & /media = external device mounting.